

# FEMMI: Finite Element Mass Map Reconstruction

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## ABSTRACT

We present FEMMI, a novel finite element method for reconstructing the projected mass density  $\kappa$  of gravitational lenses from weak-lensing shear observations. Unlike classical approaches (e.g., Kaiser–Squires), which rely on FFT-based direct inversion with periodic or Dirichlet boundary conditions, we solve the coupled Poisson problem  $\nabla^2\psi = 2\kappa$  via a hybrid finite element–boundary element (FEM–BEM) method that properly encodes the far-field behavior  $\psi \rightarrow 0$  as  $|\theta| \rightarrow \infty$ . The approach uses P3 Lagrange elements, necessary for stable computation of shear as the Hessian of  $\psi$ , and automatically selects regularisation parameters via Brent’s method on Morozov’s discrepancy principle. The forward operator  $F$  is compact and injective—crucially fixing the mass-sheet degeneracy present in all FFT methods—and admits a natural inverse-scattering interpretation through the Born approximation framework. We validate convergence on synthetic benchmarks ( $O(h^2)$  in shear,  $O(h^4)$  in potential), demonstrate effective reconstruction at 10% noise, and show that the Matérn prior enables robust inpainting of masked regions and catalog-native (unbinned) source positions. The inversion is performed via L-BFGS on the MAP objective with automatic noise-level estimation, requiring only 15–25 forward solves for  $\lambda$  selection. 

*Keywords:* [Weak gravitational lensing \(1797\)](#)

## 1. INTRODUCTION

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